

District Electrification and Exam Performance: Evidence from Uganda Using Nighttime Light Data

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Abstract—Electrification is a critical enabler of educational development, yet, its direct impact on academic performance remains underexplored. This study investigates the relationship between district-level electrification and student performance on national examinations in Uganda from 2012 to 2016. Using satellite-derived nighttime light (NTL) intensity as a proxy for electrification, we analyze its correlation with results from Uganda’s primary (PLE), lower secondary (UCE), and upper secondary (UACE) national exams. At the district level, we find a moderately strong positive correlation (Pearson coefficient = 0.517) between electrification and exam performance, with the strongest relationship at the primary level ($r=0.718$) and progressively weaker at upper secondary levels ($r=0.228$). While district-level correlations remain strong, school-level analysis reveals a weaker association ($r = 0.15$) between electrification and academic performance. This analysis merged nightlight intensity data measured within a 2 km radius of each school with corresponding exam results. This suggests that localized electrification has limited predictive power for academic outcomes at the micro level. Even in relatively well-electrified areas like central Uganda, high- and low-performing schools often sit side by side underscoring the influence of other factors such as teaching quality and socioeconomic conditions.

Notably, the gender performance gap narrows in more electrified districts ($r=-0.254$), with female students performing at par with or better than males in highly electrified regions. These findings indicate that electrification investments should be targeted to underperforming districts but must be complemented with school-specific interventions addressing teaching quality, resource availability, and gender-responsive approaches. Our research contributes to understanding infrastructure development’s role in educational outcomes in developing contexts, offering evidence-based guidance for policy interventions in Uganda and similar settings.

Index Terms—Electrification, Educational Outcomes, Nighttime Light Data, Uganda, Gender Gap.

I. INTRODUCTION

Electrification is widely acknowledged as a critical driver of educational advancement, facilitating extended study hours, improved learning environments, and the integration of modern learning technologies [1]. In Uganda, significant disparities in electrification persist, with only 7% of rural households having reliable access compared to 43% in urban areas [2]. Recent governmental initiatives to expand both grid and off-grid electrification underscore its potential to address these

disparities. Globally, electrification is associated with socioeconomic development, particularly in education [3]. However, while prior research often emphasizes enrollment or completion rates [4], the direct impact on academic performance, such as national exam results, remains underexplored. This study addresses this gap by investigating the correlation between district-level electrification, measured through satellite-derived nighttime light (NTL) intensity, and students’ performance on Uganda’s national exams. The use of NTL data provides a novel approach, offering enhanced spatial and temporal resolution compared to traditional household surveys [5].

Specifically, this research seeks to answer the following question: How does district-level electrification, as approximated by satellite-detected nighttime light intensity, correlate with students’ performance in Uganda’s national examinations (UACE, UCE, and PLE), taken at average ages 18, 16, and 12 respectively? Additionally, does this correlation vary across examination levels and differ by gender?

By exploring these dimensions, the study provides nuanced insights that could inform targeted educational and infrastructure policies aimed at mitigating regional disparities and promoting equitable educational outcomes.

II. LITERATURE REVIEW AND MOTIVATION

The relationship between electrification and educational outcomes has been extensively studied, providing a foundation and motivation for our analysis of Uganda’s national exam performance. Below, we review key studies, focusing on their methodologies, findings, limitations, and relevance to our research, which uniquely utilizes satellite-detected nighttime light (NTL) data to assess direct academic performance rather than participation metrics.

Buyinza et al. [2] explore the impact of electrification on educational outcomes in Uganda using instrumental variable probit models and data from the Uganda National Panel Survey. They report that reliable electrification significantly increases school enrollment and completion rates, particularly among female students, by reducing their domestic workload. However, the study reveals pronounced regional disparities, with electrification access limited to 7% of rural households compared to 43% in urban areas. While this study provides

valuable insights into participation metrics, it does not directly examine academic performance as measured by exam scores, which is the focus of our research. Additionally, their reliance on household survey data limits spatial and temporal granularity, a limitation our study addresses by employing NTL data for district-level analysis.

Welland [1] conducts a global review of rural school electrification initiatives, evaluating their impact on education. The report finds that electrification improves learning conditions by enabling extended teaching hours and the integration of digital tools, but it identifies challenges, including high maintenance costs, insufficient teacher training, and infrastructure sustainability issues. This broad comparative framework is useful for contextualizing our study; however, Welland’s analysis lacks specificity to the Ugandan context and does not focus on measurable academic outcomes such as exam performance. Our research addresses these gaps by focusing on Uganda and using NTL data to directly correlate electrification levels with national exam results, providing a more targeted analysis.

Koima [6] examines the short-term impact of school electrification on primary education in rural Kenya, employing a quasi-experimental design. The study finds no immediate improvements in academic performance, suggesting that factors such as teacher quality and school infrastructure may overshadow electrification’s effects in the short term. This finding underscores the potential for lagged effects, which we consider in our multi-year analysis in Uganda. Unlike Koima’s focus on primary education, our study targets both primary (PLE exams) and secondary-level outcomes (UCE and UACE exams) and employs a broader temporal scope (2012–2020) to capture potential delayed impacts, offering a more comprehensive perspective on electrification’s role in education.

Min et al. [5] investigate the use of satellite-based nighttime light imagery (DMSP-OLS) to map electrification progress in rural Africa. They confirm a strong correlation between NTL intensity and electrification but emphasize the need for localized ground-truthing data to accurately assess social impacts, such as educational outcomes. This study validates our methodological choice to use NTL data (VIIRS in our case) as a proxy for electrification. However, Min et al. focus on mapping electrification rather than linking it to specific outcomes like education, which our research extends by directly correlating NTL intensity with exam performance in Uganda.

Khandker et al. [4] analyze the welfare impacts of rural electrification in Vietnam, using household survey data and a difference-in-differences approach. They find that electrification enhances educational attainment and household welfare, but these benefits often manifest with a time lag, requiring complementary infrastructure investments to maximize impact. This insight informs our study’s design, as we use a multi-year dataset to detect potential lagged effects of electrification on exam performance. Unlike Khandker et al., who rely on survey data, our use of NTL data allows for a more scalable and spatially detailed analysis, which is particularly advantageous

in a data-scarce context like Uganda.

Dinkelman [7] studies the effects of rural grid expansion in South Africa, employing a regression discontinuity design. The study demonstrates that electrification boosts local development, including employment, which indirectly supports educational improvements through community-level changes. While Dinkelman’s focus is on economic outcomes, the implied link to education via community development is relevant to our research. Our study directly measures educational outcomes, but Dinkelman’s findings suggest that electrification’s impact on exam performance may be mediated by broader socioeconomic factors, a hypothesis we explore through control variables such as school density.

Blimpo and Cosgrove-Davies [8] provide a regional perspective on electrification in Sub-Saharan Africa, emphasizing the importance of reliability in achieving educational gains. They note that intermittent power supply often undermines the potential benefits of electrification in schools, a challenge prevalent in Uganda. This study complements our research by highlighting the quality of electrification as a critical factor, which NTL data may not fully capture. We address this limitation by acknowledging that NTL intensity reflects access rather than reliability, and we recommend future research to incorporate ground-level data on power stability.

Grimm et al. [9] investigate the impact of electrification on educational outcomes in rural India, using a panel dataset and regression analysis. They find mixed results, with electrification improving study hours but not consistently translating to better academic performance, possibly due to confounding factors such as parental education. This study contrasts with Koima [6] by suggesting some positive effects, albeit indirect. Our research builds on Grimm et al. by focusing on a direct outcome (exam scores) and using NTL data to overcome the data limitations often faced in household surveys, providing a more robust analysis for Uganda.

Finally, Henderson et al. [10] pioneer the use of NTL data to measure economic activity, demonstrating that light intensity correlates with GDP growth. Their methodology underpins our use of NTL as a proxy for electrification, as economic development often accompanies infrastructure improvements like electricity provision. While Henderson et al. focus on economic outcomes, their validation of NTL data as a reliable proxy supports our approach to linking electrification with educational outcomes in a developing country context.

Our study contributes to this literature by combining NTL data with district-level exam performance in Uganda, offering a novel perspective on the electrification-education nexus. By focusing on academic outcomes rather than participation metrics and using a scalable, satellite-based approach, we address gaps in the existing research and provide actionable insights for policy.

III. METHODOLOGY

A. Data Collection

This study utilized publicly available data sources to examine the relationship between electrification and educational

performance in Uganda. The primary datasets used include national examination results and satellite-derived nighttime light radiance data.

1) *National Examination Data:* The Uganda National Examinations Board administers three major national examinations that facilitate the promotion of students through the three major schooling tiers. The Primary Leaving Examinations (PLE) that is used to promote students from primary school to lower secondary school, the Uganda Certificate of Education (UCE) examinations that is used to promote students from lower secondary school to upper secondary or high school and finally the Uganda Advanced Certificate of Education (UACE) that is used for national University placement and selections. The results from these examinations are not ordinarily published in raw form however limited datasets are released and made available to researchers through other repositories where they can be obtained for further analysis. We obtained the examination results from DataDotUG [11]. These included:

- Primary Leaving Examinations (PLE): 2010–2015
- Uganda Certificate of Education (UCE): 2012–2016
- Uganda Advanced Certificate of Education (UACE): 2011–2015

The examination datasets sourced were organized at the **school-year level**, with each record representing the performance of a single school in a specific examination year. The structure of the dataset is explained in Table I.

TABLE I
SCHOOL-YEAR-LEVEL EXAMINATION DATA SET

Variable	Description
Identifiers	
YEAR	Exam year
DISTRICT	School district
SCHOOL	School name
Candidates	
CANDIDATES	Total candidates
FEMALE CANDIDATES	Total female candidates
MALE CANDIDATES	Total male candidates
Performance	
AGGREGATED PERFORMANCE	Overall school performance score
FEMALE PERFORMANCE	Female performance score
MALE PERFORMANCE	Male performance score

This structure facilitated both district-level aggregation (for broader regional analysis) and school-level spatial analysis (for localized electrification-performance correlation)

2) *Nightlight Data:* To estimate electrification levels across Uganda, we used nighttime light radiance data derived from the Visible Infrared Imaging Radiometer Suite (VIIRS) operated by the National Oceanic and Atmospheric Administration (NOAA) [12].

We specifically used the VIIRS Day/Night Band (DNB) annual composite products. These datasets capture low-light imaging at night and measure the intensity of visible and near-infrared light reflected or emitted from the Earth’s surface. The DNB is sensitive to artificial lighting, making it a widely accepted proxy for electrification, particularly in studies of

infrastructure and development in low- and middle-income countries.

To align with administrative boundaries and facilitate district-level analysis, we used the Global Administrative Unit Layers (GAUL) 2015 – Level 1 shapefile from the Food and Agriculture Organization (FAO). Uganda’s 112 districts were delineated based on this shapefile.

Annual average radiance values were calculated for each district using zonal statistics, which involved overlaying the district boundaries onto the raster radiance images and computing the mean radiance value for each zone.

To enable spatial analysis at the school level, geographic coordinates for schools were extracted from OpenStreetMap (OSM) using a search based on a combined string comprised of the school name and district name as retrieved from the examinations dataset. Each school’s location was matched with surrounding nightlight intensity, measured within a 2-kilometer radius, to evaluate localized electrification effects.

B. Data Processing

1) *Pre-Processing:* To prepare the UACE data for analysis, in addition to zeroing empty numerical fields and checking for duplicates, schools that did not have records reported in every one of the years studied were removed from the dataset.

Schools in Uganda are not required to have unique names, they are given a unique registration number by the ministry of education at registration but these numbers were not included in the dataset we were working with. Fortunately while schools are not required to have unique names, the regulations around naming make the existence of duplicate names unlikely. However since the risk remains we removed duplicate name and district combinations from the dataset. This had little effect on the eventual analysis as possible duplicates likely from how school names were reported were less than 0.1% of the dataset.

2) *Ranking of Schools and Students:* The data contains the total number of students in a school and the total number of students in various performance divisions or groupings. We needed to create a comprehensive scoring and ranking algorithm that would allow us to rank the schools and districts and analyse the best and worst schools and districts as needed.

To enable meaningful comparison of PLE and UCE performance across districts, we computed an adjusted performance score using a Bayesian-inspired smoothing technique. First, we estimated the absolute number of Divisions for each school by multiplying the percentage of Divisions’ students with the total number of candidates. These values were then aggregated by district and year. To account for volatility in districts with fewer candidates, we applied a smoothing formula where each district’s score was adjusted toward the national average.

The resulting adjusted scores were normalized to a 0–1 scale to facilitate fair comparisons. Finally, for each district, we computed the average of scores across all available years (2010–2015) to obtain a single, comparable performance metric.

For UACE, a different approach was used that involved ranking the schools in order of the percentage of students they

had in the top 3 performance rankings; 11-15 points, 16-20 points and 21-25 points. Where 25 points is the highest score possible and represents the best possible performance for a student.

With both techniques we were able to create a ranked list of schools and districts based on performance.

3) *Extraction and Analysis of Nighttime Light Data:* In the district-level approach, nighttime light data from VIIRS monthly composites (2012–2016) were aggregated to analyze district-level patterns in Uganda’s 112 administrative districts. Administrative boundaries were obtained from the FAO GAUL 2015 dataset, and annual mean radiance values were calculated by spatially averaging the VIIRS avg_rad band within each district boundary. This approach linked annual nighttime light intensity a proxy for electrification to district-level exam performance, enabling an assessment of whether broader access to electricity correlates with educational outcomes.

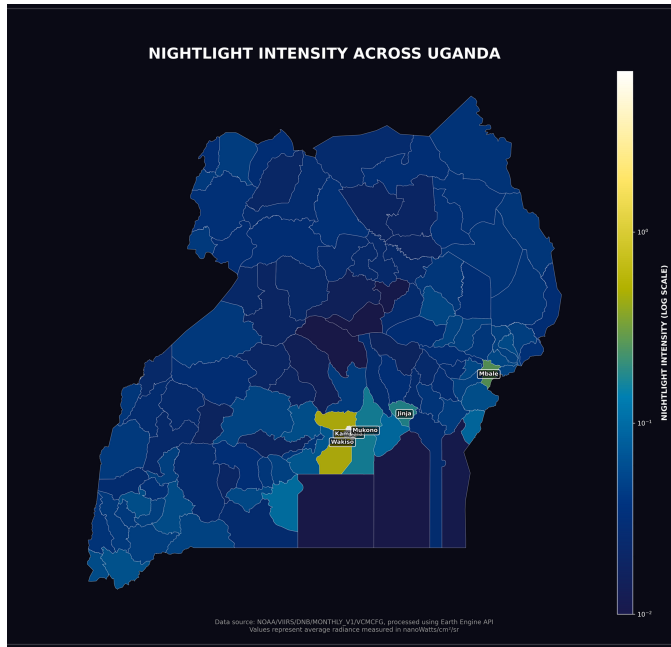


Fig. 1. Nightlight Intensity District Analysis

On a School-Level approach:
School locations were identified using names from the Uganda National Panel Survey, with coordinates georeferenced via OpenStreetMap, resulting in 1,955 schools. Nightlight intensity was extracted annually (2012–2016) using a 2-km circular buffer around each school. The VIIRS avg_rad band was averaged annually to minimize seasonal variations, capturing localized radiance as a measure of electrification near schools. The 2-km buffer was selected to approximate the spatial scale of potential electrification effects on school communities, though sensitivity to buffer size warrants further investigation. This method tested whether proximity to electrified areas, inferred from nighttime light intensity, correlates with school-specific exam performance.

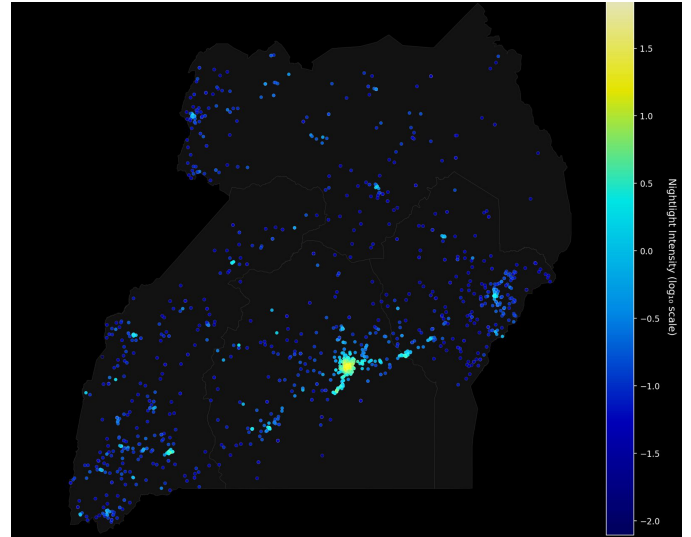


Fig. 2. Nightlight Intensity School Analysis

IV. FINDINGS

After merging district-level nighttime light intensity data with our calculated school exam performance index, we analyzed whether any correlation existed between the two variables.

Our analysis revealed a Pearson coefficient of 0.517, indicating a moderately strong correlation between electrification and national exam performance. Districts such as Kampala and Wakiso, which exhibit high nighttime light intensity, consistently ranked among the top performers in national examinations. However, this pattern did not hold universally. For example, Mbale, a district with relatively high electrification compared to other regions in Uganda was among the poorest-performing districts academically. This divergence suggests that while electrification may play a role in educational outcomes, it is likely one of many contributing factors. Socioeconomic conditions, resource allocation, and regional disparities in teaching quality could also significantly influence exam performance.

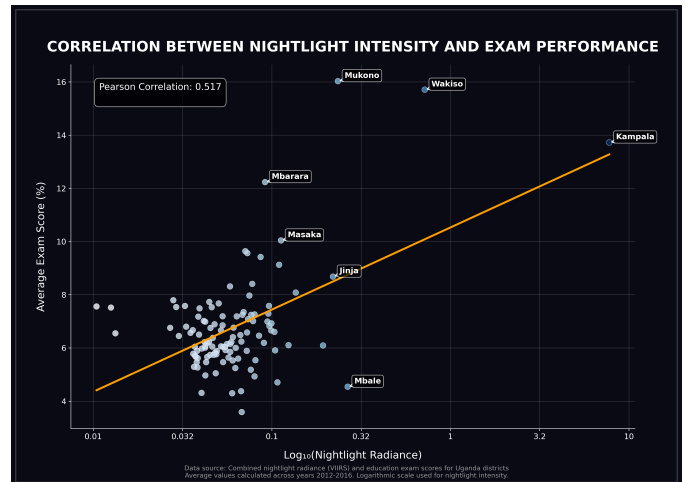


Fig. 3. Nightlight Intensity against District level examination Performance

To ensure the correlation persisted over time, we conducted additional longitudinal analysis by mapping each district's exam performance across multiple years. This allowed us to identify patterns in the relationship between electrification and academic outcomes. As expected, the top three performing districts consistently aligned with the three most electrified regions in Uganda. Notably, the gap in exam performance between these top districts and the rest of the country was substantial, as illustrated in the figure below.

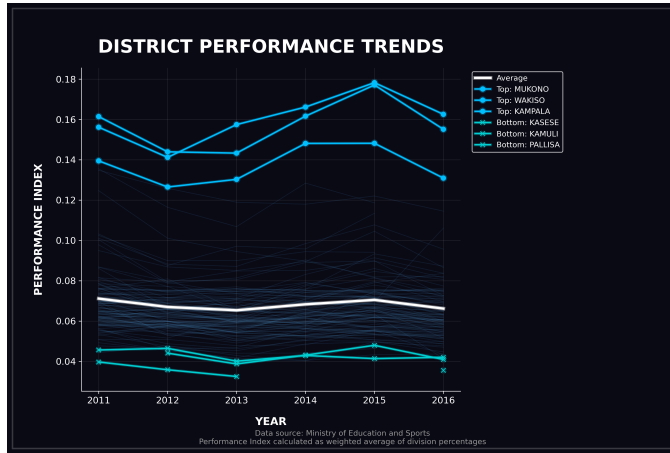


Fig. 4. Nightlight Intensity against District level examination Performance

Overall, our district-level analysis revealed a strong correlation (Pearson = 0.517) between electrification levels and national school examination performance in Uganda. To test whether this relationship held at a more granular scale, we shifted from a district-level to a school-level analysis.

For the school-level assessment, we merged nightlight intensity data (captured within a 2km radius of each school) with individual schools' exam performance. Contrary to our initial assumptions, we observed a weak Pearson correlation coefficient of 0.15, suggesting no meaningful link between localized electrification and academic outcomes. This finding was surprising, as most schools are clustered in central Uganda, a region with relatively high electrification. Yet even here, high-performing schools were often located adjacent to low-performing ones, highlighting the role of additional factors like teaching quality or socioeconomic conditions in shaping results.

Despite the weak overall correlation, nearly all schools with low electrification performed below average, while almost all top performers had high electrification. This suggests that while electrification alone does not guarantee strong exam results, it may act as a foundational requirement.

Next, we examined whether the correlation between electrification and academic performance varied across different educational stages. To do this, we analyzed three national exams: the Primary Leaving Examination (PLE), Uganda Certificate of Education (UCE), and Uganda Advanced Certificate of Education (UACE). Using the same methodology as earlier, we

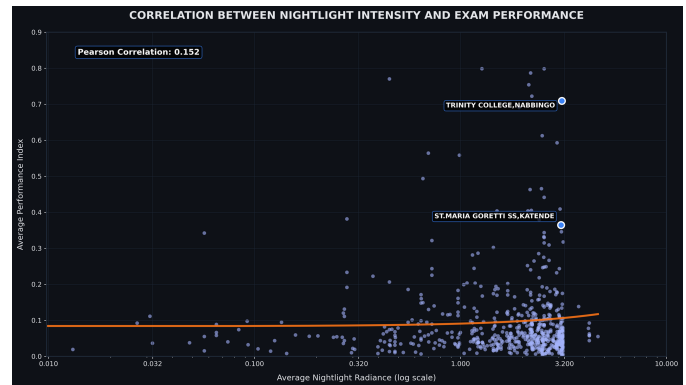


Fig. 5. Nightlight Intensity against School Performance Analysis

calculated district-level exam performance indices and merged these with averaged nightlight intensity data for each district.

We calculated the Pearson correlation coefficients for all three exams against nightlight intensity. The results showed scores of 0.718 (PLE), 0.648 (UCE), and 0.228 (UACE). As expected, the correlation persists across all exams, but it decreases steadily as students progress through their education from primary (PLE) to advanced (UACE) levels. Notably, while primary-level performance shows a stronger link to electrification, advanced exams like the UACE correlate far less. This divergence may reflect the growing complexity of educational needs at higher levels or the rising influence of non-infrastructure factors



Fig. 6. Nightlight Intensity against PLE, UCE and UACE performance

we then studied the correlation between electrification and national examination performance across genders. Using the same technique as before, we calculated performance indices separately for each gender, grouped the results by district, and merged them with the corresponding average nightlight intensity data per district. As expected, we found strong Pearson correlation coefficients for both genders 0.525 for females and 0.517 for males indicating a clear positive relationship between electrification levels and academic performance.

We then computed the gender gap in academic performance

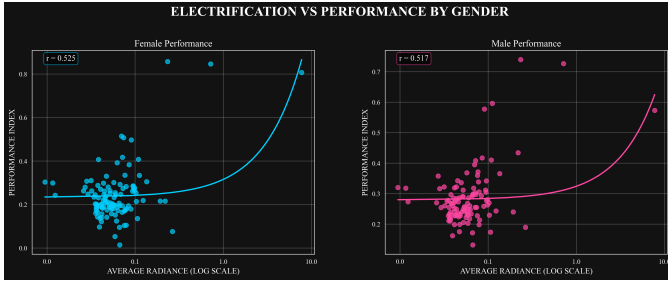


Fig. 7. Male Vs Female Examination performance against average Night light intensity

by subtracting the female performance index from the male performance index for each district and calculated the Pearson correlation coefficient with respect to nightlight intensity. We found a value of -0.254, which suggests that the performance gap between males and females tends to narrow as districts become more electrified.

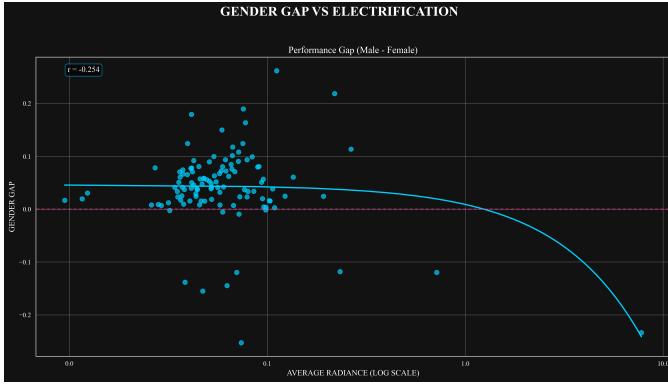


Fig. 8. Male - Female Examination performance gap against average Night light intensity

Finally, we looked into the gender composition of the top 20 and bottom 20 districts based on national examination performance. We observed that in the high-performing districts which also tend to be highly electrified, female students excelled or performed on par with their male counterparts. On the other hand, in many of the bottom 20 performing districts, males outperformed females by substantial margins.

V. DISCUSSIONS

This research analyzed the relationship between district-level electrification, determined by nighttime light intensity (NTL), and national examination performance at primary (PLE), lower secondary (UCE) and upper secondary (UACE) levels in Uganda. The results showed a consistent trend where districts with higher NTL values performed better on national exams.

This connection is consistent between different age groups and educational levels, suggesting that access to electricity through improvements in school facilities or in the broader community could help support the achievement of better educational outcomes. However, the relationship does not



Fig. 9. Top 20 vs Bottom 20 Districts by performance

appear to be linear at the school level, where there are high-performing schools in areas of low nightlight radiance. These outliers suggest that even though electricity can facilitate the improvement of learning environments, other factors such as school leadership, available resources, or student backgrounds can also play an important role in shaping academic performance.

Limitations

Despite these insights, several limitations influenced the scope and confidence of the analysis. One major challenge was inconsistent and incomplete reporting of school-level exam data. Missing years and duplicated school names affected the temporal reliability of some performance trends. Similarly, while OpenStreetMap data enabled geolocation on the school-level, the nightlight satellite data lacked the spatial granularity needed to precisely assess lighting conditions at each school. Rural areas, often home to high-performing boarding schools, exhibited low NTL values even when electrified, due to broader radiance limitations of satellite measurements in sparsely populated zones. Additionally, the examination data covered only five years which is a too narrow window to capture longer-term effects of electrification on educational outcomes.

Another limitation is that the NTL data may capture electrification but does not reflect the reliability of the electricity at the different locations.

Policy Implications

These findings suggest clear policy implications. Electrification strategies should be data-driven, prioritizing districts with both low performance and low NTL values. However, infrastructure alone is insufficient for school-specific interventions, including quality teaching, student resources, and support systems, to accompany energy investments. Verification of satellite data with local knowledge is vital to ensure interventions are accurately targeted. Policymakers should also plan for the short-term education outcome advancements in cases where electrification is planned to improve the education outcome since its results take time to reflect, and differentiated approaches are needed for each level of education.

Moreover, differentiated approaches are essential, as the correlation between electrification and performance varies by education level with strongest in primary and weakest in upper secondary, which indicates that infrastructure improvements must be complemented with more specialized resources at higher levels. In addition, gender-responsive policies are critical. In well-electrified districts, gender gaps narrow or even reverse in favor of girls, while in poorly electrified areas, girls fall significantly behind. This underscores the importance of electrification policies that consider the unique needs of female students, such as reducing domestic burdens and ensuring access to study environments in the evening.

Future work

Future researchers should explore how electrification affects pedagogy, such as extended study hours or technology adoption. Also analyzing differences between day and boarding schools and how electrification impacts each could help in getting deeper insights into how students might benefit from improved energy access in both school and home settings.

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